multimedia creation

Tools + Resources

Technology ≠ Fancy gadgets

It’s a creative right and a way of expression.

Who am i?

I have a ***Computer Science*** background: programming, networking, databases, graphics.

→ ***Computation Arts*** to mix tech + art.

I work at the Society of Arts and Technology (sat.qc.ca) in Montréal, Canada as a multimedia programmer & designer.

What is multimedia art?

Input → rules → output

images: camera, mic, sensors, data,

if loud, make the circle bigger; if bright, change color.

Outputs: screen, sound, light, motors, movement, or the web

idea generator: randommedia.art

multimedia creation tools

VISUAL PROGRAMMING

MaxMsp/Jitter/MaxforLive
Vvvv
TouchDesigner
Isadora
Notch (?)
Pure Data (Pd) / plugDATA
Ossia Score
Smode
Blender
Unreal engine, Unity, Godot
Houdini

CREATIVE CODING

Processing
p5.js
openFrameworks (c++)
Unreal Engine
Cinder
Shadertoy
Hydra
Html / css/ javascript/ gsap3
supercollider

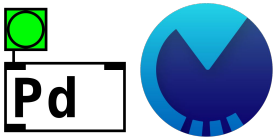
SOUND DESIGN

Ableton live
Reaper
spatGRIS
VCV Rack
Supercollider
Bitwig

HARDWARE /INTEGRATION

Sensors
Immersive capture
Robotics
Arduino, esp32, Raspberry Pi
MadMapper
Point Cloud input
Game controllers
MIDI controllers
LeapMotion
OBS

multimedia creation tools my toolbox



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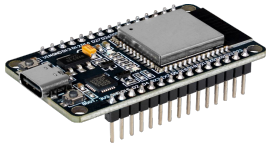

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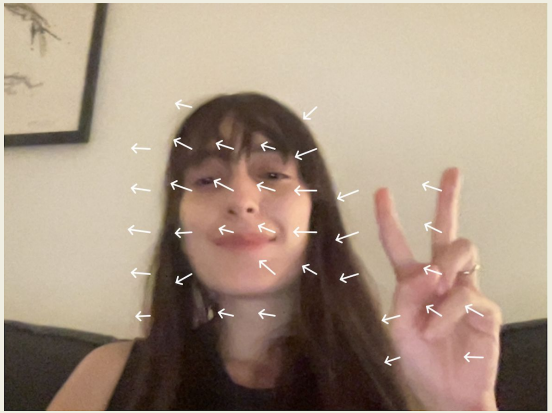
→ forums, github repositories, discord servers.

BASIC EXAMPLES

→ webcam input
→ motion detection
→ create visuals

https://editor.p5js.org/kardakaris.k/sketches/eTISz1-3Q



→ microphone input
→ analysis
→ create visuals

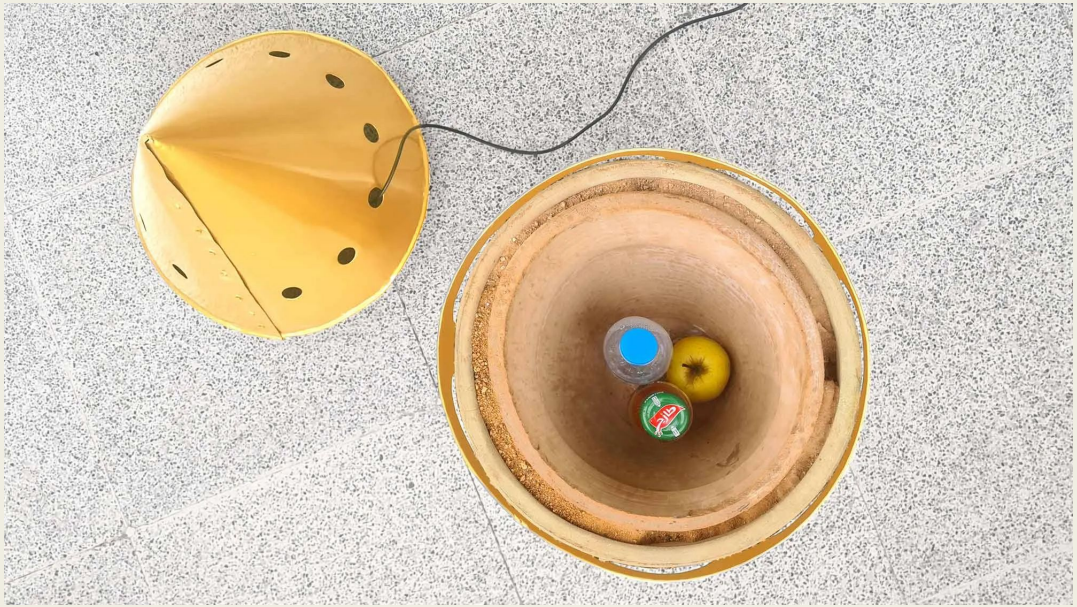
p5.js or TouchDesigner
therewasaguy.github.io/p5-music-viz

→ midi controller
→ parameter mapping
→ beautiful shader

(live ossia score demo! ++)
control shaders with any value input

Selected works by Syrian diaspora

1. *Revolution Refridge* (low-cost / DIY & community tech)
   ars.electronica.art/starts-prize/en/revolution-refridge



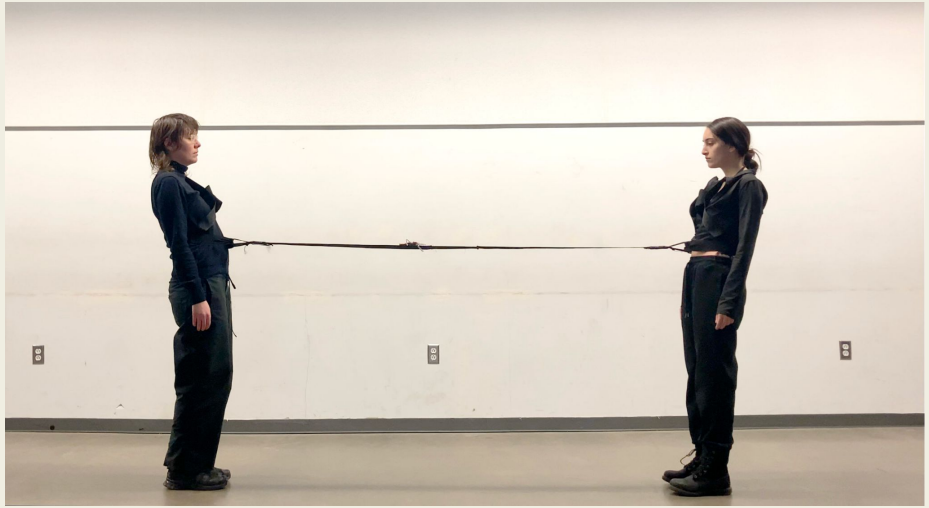
Selected works by Syrian diaspora

1. An effort to reconstruct the ancient city of Palmyra as an immersive virtual environment, in a way to immortalize the monuments. en.wikipedia.org/wiki/NEWPALMYRA

Selected works by Syrian diaspora

- Acoustic Investigation into the Syrian Regime Prison - Lawrence Abu Hamdan
- khaledbarakeh.com/sp/design-of-necessity
- sulafahijazi.com
- .... To be continued on Discord!

Personal works



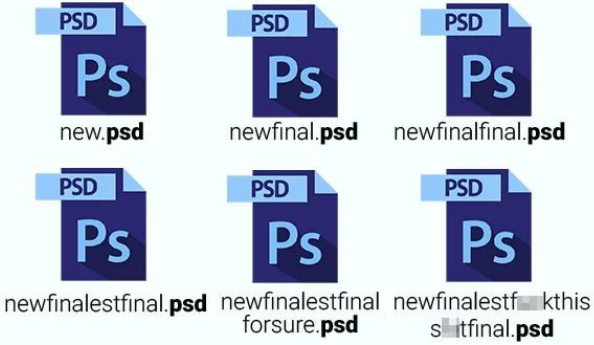
Being-With
A Sound Performance for Motion and Stretch Sensors

An all-time glow
Re-Examining Light and Noise Pollution as an Aesthetically and Conceptually Charged Mediascape

Archiving / documentation

Document each step of your work and your process.
What materials you're using? Inputs? Outputs?

Keep versions (for code you can use git)



Explore: **online resources** & **archives**

Open-access academia (FREE)

- **arxiv.org**, with subsections for different domains
  cs.HC : Human-Computer Interaction

Curated academia
(behind paywalls but abstracts are available)

- ISEA (International Symposium on Electronic Arts) https://isea-archives.org
- Siggraphhttps://s2024.siggraph.org/program/art-gallery
- NIME (New Interfaces for Musical Expression) https://nime.org/archives/
- ACM https://dl.acm.org

Big big festivals

- Ars Electronica, ZKM, CTM, MUTEK,..

Learning

- Kadenze.com, openprocessing.org
- art21.org/medium/digital-media/
- creativeapplications.net
- rhizome.org/artbase

Hardware projects

hackster.io instructables.com hackaday.io

Join our small community

Questions?

Merci / Thank you / شكراً